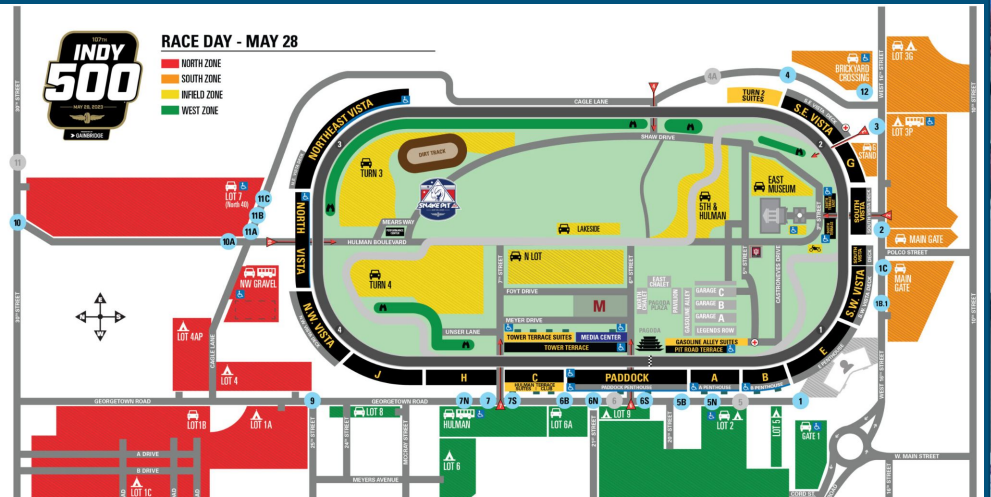
Shell Eco-Marathon

Jano Vasquez-Jaffe & Kimaru Boruett

Shell Eco-Marathon

- **Worldwide Engineering Competition**
  - WAY less fast and money intensive version of Formula 1
- **Multiple Categories**: Prototype and Urban
- **Different Powertrain Systems: Hydrogen**, ICE-Hybrids, & Battery EVs







Prior Vehicles


2024 Test Drive

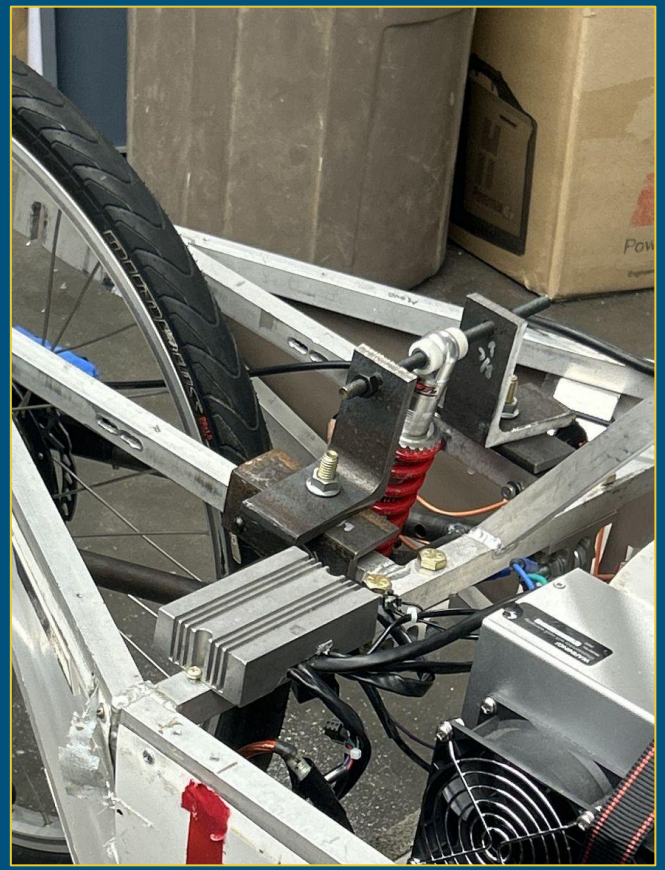


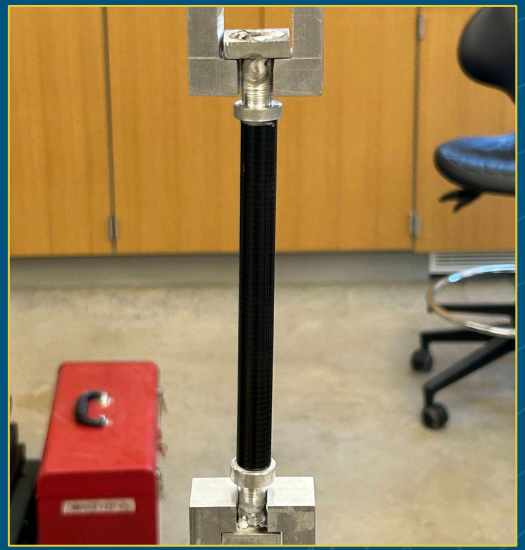
2015 Competition Team


2018 Front Suspension


2012 - Original Chassis Construction

Past 3 Years

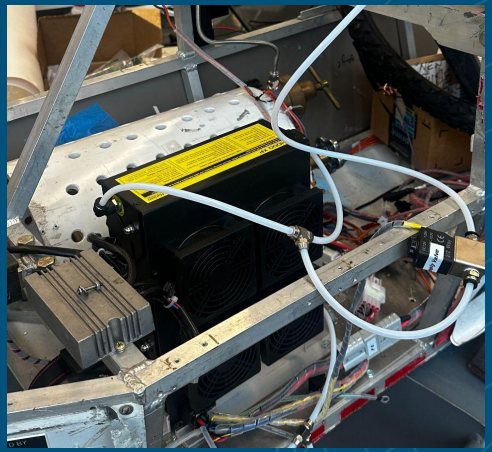






Current Progress







Motivation

We need a telemetry system that is simple to operate, fast to iterate, and tailored to the signals that matter.

- Competition software introduces overhead: setup complexity, extra dependencies, and workflow friction for a student team.
- The car produces raw data streams that still need decoding and interpretation before they become useful engineering information.
- We need one pipeline that supports development now with simulation data and then transitions cleanly to live sensors already installed on the car.

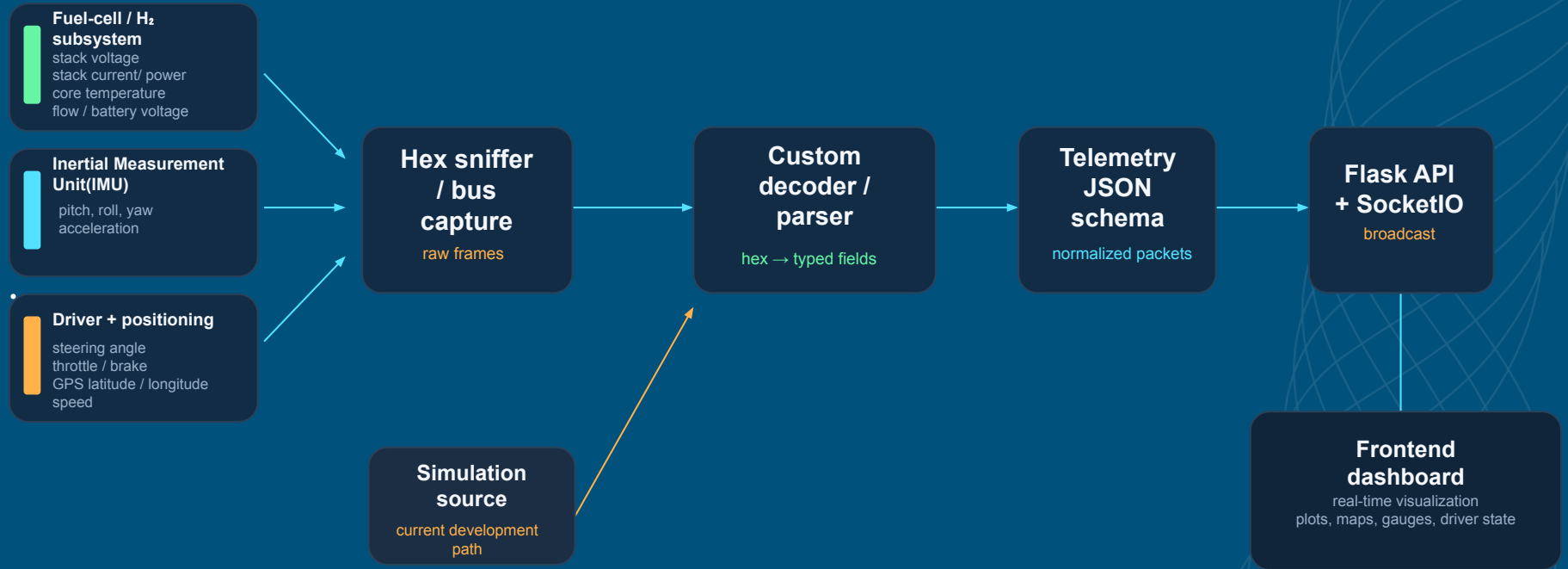
Design requirements

Low overhead	No unnecessary platform complexity — just capture, decode, process, and display.
Structured telemetry	Convert raw hex data into consistent fields for fuel-cell health, dynamics, driver input, and position.
Real-time streaming	Push updates quickly enough for a live dashboard and future testing sessions.
Sensor-ready architecture	Keep the transport and frontend stable so only the input layer changes when real sensors are connected.

Motivation Summary: Build a lean telemetry workflow the team can control end-to-end.

Telemetry Flow: Car to Dashboard

Opening systems view: what signals we capture and how they move through the stack



Key idea: we are replacing heavyweight competition software with a lighter pipeline that reads vehicle data from a hex sniffer, decodes it, and streams only the telemetry we need.

What we've done so far

Frontend / UI

- Spent most of the project time ironing out the frontend and making the dashboard readable in real time.
- Set up pages for the main live view and plots.
- Verified that connected clients can subscribe to a live socket stream and update continuously.

Data model

- Defined a structured telemetry schema for H_2 status, dynamics, IMU data, driver inputs, GPS, and weather.
- Used simulation payloads so the frontend and backend could be exercised before full sensor hookup.
- Built in track-coordinate loading so motion can be replayed over a real course.

Parsing

- Worked through the hex-accessed data using a serial sniffer (serialTool) and decoded log format.
- Mapped raw fields into engineering quantities like stack temperature, voltage, current, and power.

Pipeline

- Avoided the competition software overhead.
- Set up the custom route: hex sniffer → custom script → API → dashboard.

Current Progress: Main Page



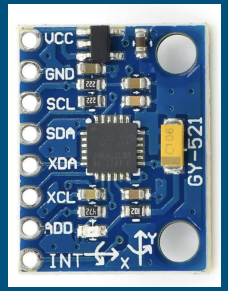
Current status: the software path is proven with simulation and decoded logs; the main unfinished step is swapping the input source from replayed data to live sensor feeds.

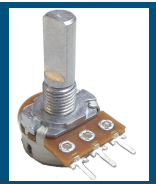
Current Progress: Hydrogen Stack Data



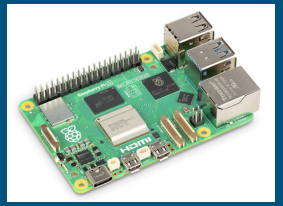
What work remains to be completed

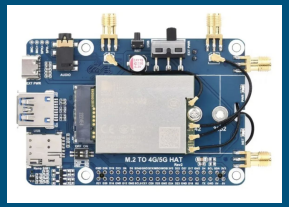
1. Connect the real sensors



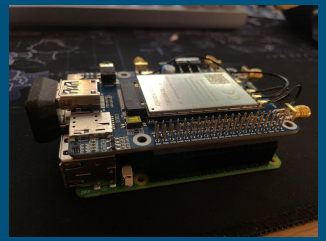


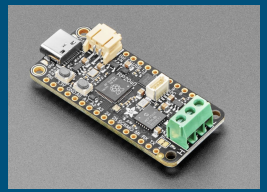
2. Finalize the hex-sniffer interface





3. Validate



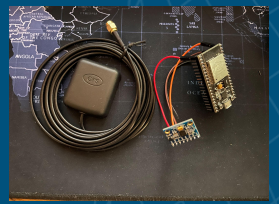


4. Test on the vehicle





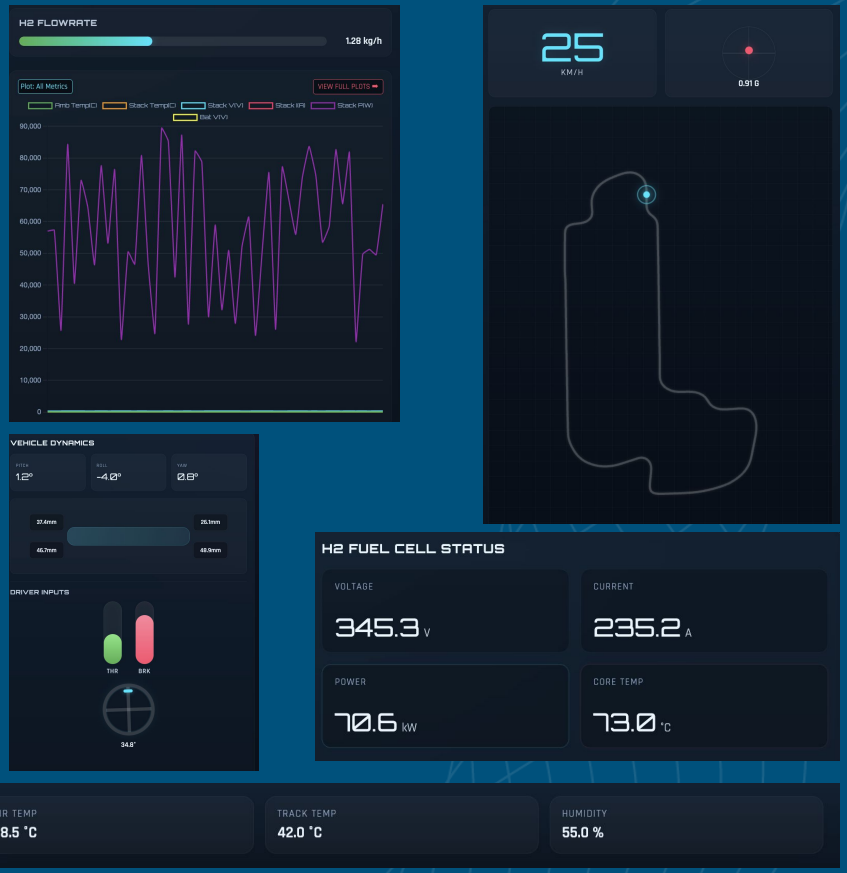




End goal: a lightweight, team-controlled telemetry system that reads live car data directly, avoids competition software overhead, and gives a reliable real-time view of vehicle health and driver state.

Challenges

- Setting up a development environment for our Adafruit RP2040 CAN bus development board
- Getting MPU6050 IMU libraries that directly work with our board
- Setting up a real-time database to capture the data



Preparation for Race Day

- Finish moving power train to new vehicle
- Add tires
- Perform mock inspection

This Year’s Race: Logistics

- April 7-11th
- Vehicle Transport by van, others by air
- Rotating van drivers/ air travelers
- Hotel near track
- Meals paid for by OSE